Improved Modeling for Moving Sources of Radio Frequency Interference and Impact on Flagging Strategies

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Mise en scène

The **Epoch of Reionization** (EoR), characterized by the birth of the Universe's first stars and galaxies, represents a rich and informative period of cosmic history. A distinguishing feature of the EoR is its abundance of **neutral hydrogen**, whose hyperfine spin-flip transition emits radiation at a rest-frame wavelength of 21 centimeters. The growing field of **21-cm cosmology** aims to map early cosmic periods using neutral hydrogen, opening a direct observational window onto early structure formation [1–3].

The task is arduous. The 21-cm signal is incredibly *faint* relative to radio foregrounds. In particular, **radio frequency interference** (RFI), often several *orders of magnitude brighter* than astronomical foregrounds—themselves several orders of magnitude brighter than the 21-cm signal—represents a significant experimental obstacle and a major focus for current research groups.

Radio frequency interference (RFI)

RFI arises from many different sources. From satellite emissions [4,5] to lightning strikes [6] and even *reflections* of ground-based emissions off of aircraft flying overhead [7,8], **RFI is difficult to characterize, predict, and remove**.

Current mitigation strategies focus on **identifying and excising RFI**. They typically inspect interferometric data in **time–frequency space**, often visualized as a **waterfall plot**. In a waterfall plot, RFI shows up as a bright rectangle or **streak** (see top row of Fig. 1), which can be algorithmically identified and **flagged**.

This strategy has obvious limitations: getting rid of any RFI-contaminated time/frequency bins *also* gets rid of the underlying 21-cm signal, **reducing the overall amount of usable data**. For experiments requiring $\mathcal{O}(1000)$ hours of clean data to achieve a detection, excessive RFI flagging quickly becomes a serious obstacle.

Characterizing RFI

Although many sources of RFI are difficult to model, ***moving* sources of RFI** can leave a coherent geometric signature in interferometric data. The key idea comes from the basic measurement equation of interferometry: the **Fourier transform**.

In the flat-sky approximation, the Fourier transform relates the sky brightness distribution, defined in terms of the direction cosines (ℓ, m), to the observed *visibilities*, defined in terms of ground-based (u, v) coordinates [9].

$$F(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\ell,m)e^{-j2\pi(u\ell+vm)}\,d\ell\,dm. \qquad (1)$$

Stationary RFI

A compact unresolved source appears as a **fixed** "dot" on the sky, which we can model using a **Dirac delta function**. The Fourier transform of a Dirac delta function is **constant** in amplitude, leading to the following interpretation:

- Each antenna-pair correlation, or **baseline**, samples one point in the uv-plane at a given time and frequency. For an unresolved source, these samples have approximately the same visibility amplitude.
- If each sampled uv-coordinate is colored by its measured visibility amplitude, the resulting amplitude map should appear approximately **uniform** (see bottom-right panel of Fig. 1).

Moving RFI

A *moving* source of RFI, however, is better modeled using a **streak** (Eq. 2). The Fourier transform of a streak works out to a product of two **sinc** functions (where $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$), modulated by a phase term (Eq. 3).

Important equations

Streak model: a rectangular source of length r, thickness τ, rotated by angle θ:

$$f(\ell,m) = \begin{cases} 1, & |x_\parallel| \le r/2, \quad |x_\perp| \le \tau/2, \\ 0, & \text{otherwise,} \end{cases} \qquad (2)$$

where:

$$x_\parallel = (\ell-\ell_0)\cos\theta + (m-m_0)\sin\theta,$$
$$x_\perp = -(\ell-\ell_0)\sin\theta + (m-m_0)\cos\theta.$$

Fourier response: a product of two *sinc* functions, rotated by the same angle:

$$F(u,v) = r\tau\ \mathrm{sinc}(ru_\parallel)\ \mathrm{sinc}(\tau u_\perp)\,e^{-i2\pi(u\ell_0+vm_0)}, \qquad (3)$$

where:

$$u_\parallel = u\cos\theta + v\sin\theta,$$
$$u_\perp = -u\sin\theta + v\cos\theta.$$

Observations

Our analysis focuses on data from the **Murchison Widefield Array** (MWA) [10,11], a radio interferometer located in Western Australia. More broadly, the results and conclusions presented here are relevant to **low-frequency interferometric experiments searching for the faint 21-cm power spectrum**.

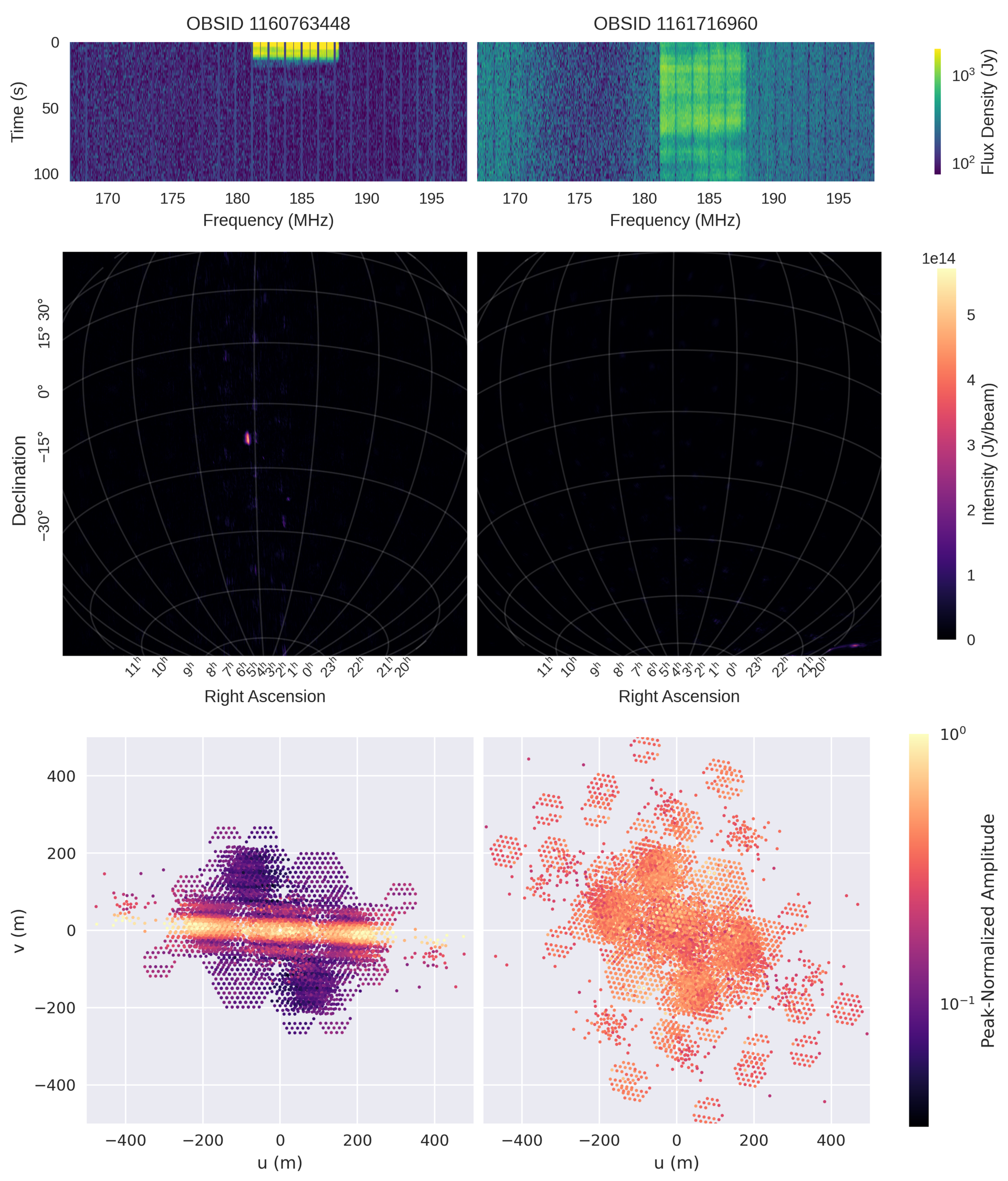

Figure 1. **Top:** waterfall plots for two MWA observations containing RFI. **Middle:** images of the most contaminated time integrations. The reference observation, MWA OBSID 1160763448, resolves a **streak-like source** near the phase centre. **Bottom:** visibility amplitudes plotted in the uv-plane for the contaminated time and frequency range. The moving source produces a clear **sinc-like pattern**, while the unresolved source produces an approximately uniform amplitude response.

Why this matters

Accurate models of moving RFI could help shift current **flag-and-exclude** strategies to a **model-and-subtract** approach, preserving a higher fraction of usable data while reducing contamination.

Recent work has begun to **localize** RFI [8] and **subtract** it [12,13], but further modeling accuracy is needed to meet the **stringent precision requirements** of 21-cm EoR measurements.

Comparison with simulations

To validate the analytical model, we compare it against **numerical simulations**. Real observations contain thermal noise, calibration residuals, and astrophysical sky emission, so a controlled simulation provides a cleaner test of our model.

Using the `pyuvsim` [14] Python package, we simulate a moving source of RFI from a set of trajectory points in right ascension and declination. To make the simulation realistic, we **track the real RFI source** in our reference observation using the `WSClean` [15] imaging software, and use the measured image-plane trajectory to define our simulation parameters.

Right ascension and declination are converted to ℓ and m direction cosines, which we can then use to compute the analytical solution directly from Eq. 3.

As Fig. 2 shows, the analytical model **accurately reproduces** the same sinc-like uv-plane morphology seen in the data and in the `pyuvsim` simulation.

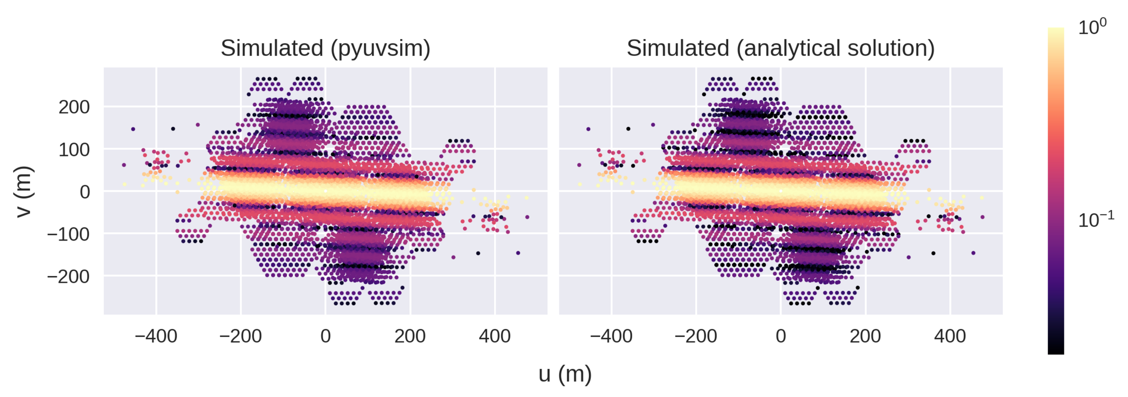

Figure 2. Comparison between the `pyuvsim` simulation and the analytical Fourier model from Eq. 3 for the reference RFI event.

Moving RFI is baseline-dependent

A moving source of RFI produces a sinc-like pattern in the uv-plane, so **different baselines can observe very different RFI amplitudes**. Fig. 3 illustrates this: baselines falling on bright lobes of the pattern show strong RFI features in their waterfall plots, while baselines near suppressed nulls show little or no detectable contamination. This creates a challenge for **per-baseline flagging**: the same RFI event may be obvious on some baselines but nearly invisible on others.

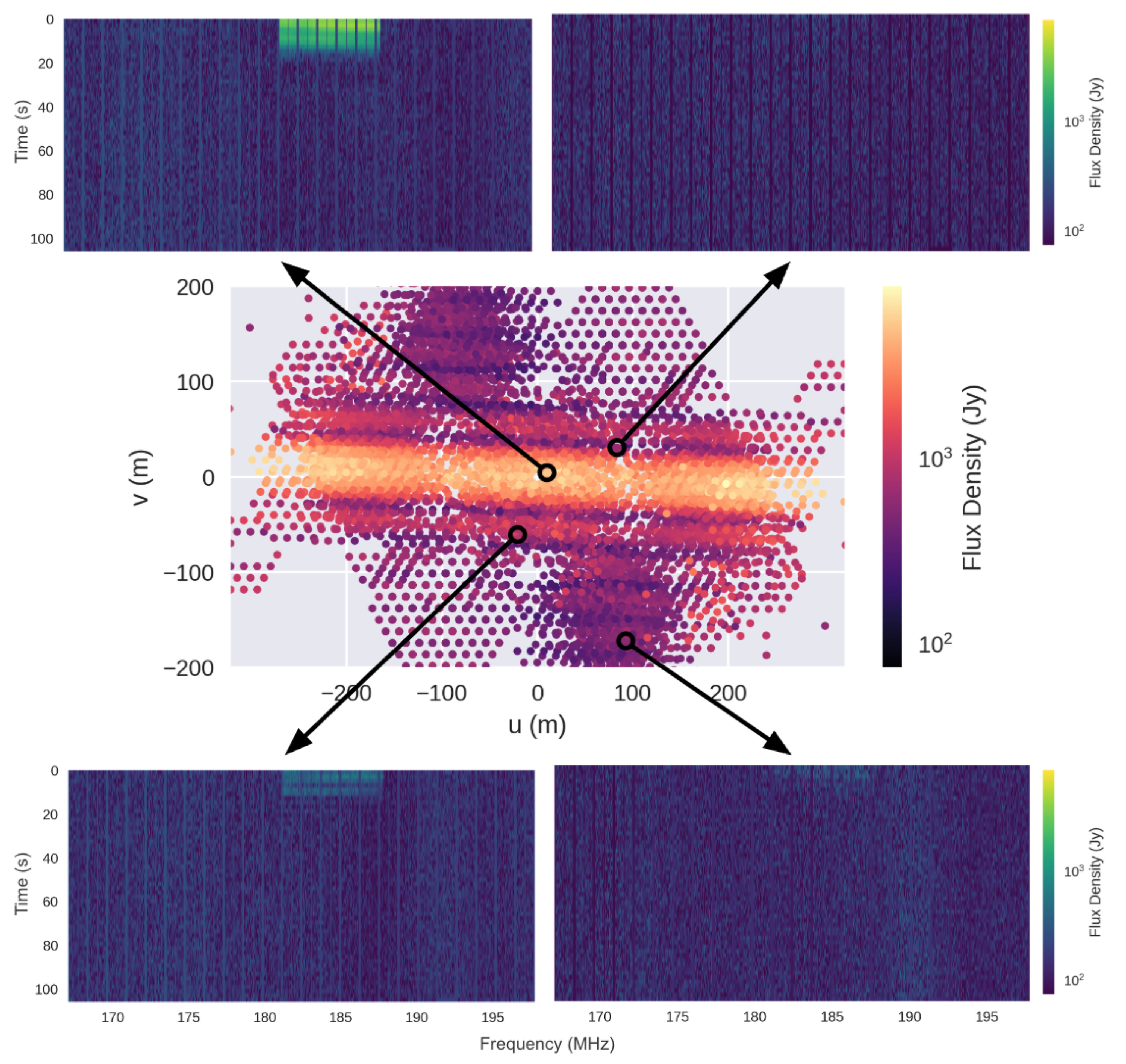

Figure 3. Waterfall plots for four randomly selected baselines of our reference observation are shown in the four corners. The central panel shows each baseline's uv-coordinates and its average visibility amplitude across the contaminated frequency range for the first time integration.

Impact on flagging strategies

An immediate consequence of the sinc-like uv-plane response is that per-baseline flagging can miss RFI on suppressed uv-modes, **allowing faint contamination to remain undetected**.

To illustrate this, we apply three different flagging algorithms to the reference observation: `AOFlagger` [16], `SSINS` [17], and χ^2 [18]. While `AOFlagger` inspects individual baselines and generates baseline-dependent flag masks, `SSINS` and χ^2 use baseline-aggregated summary statistics to produce a **single, global flag mask** that can be applied to all baselines.

Fig. 4 shows that while `SSINS` and χ^2 are able to identify the main contaminated time-frequency region, `AOFlagger` captures the strongest contamination only on a subset of the baselines.

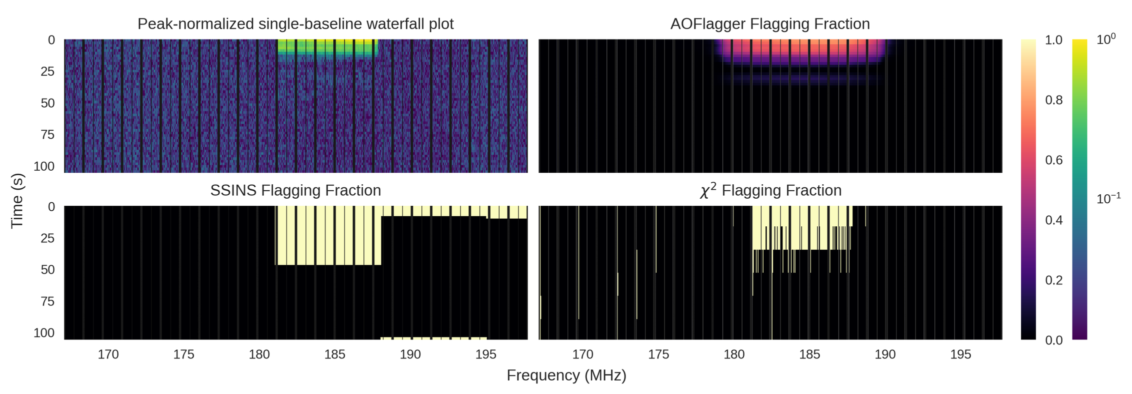

Figure 4. Top left: peak-normalized single-baseline waterfall plot for our reference observation. Top right: `AOFlagger` flagging fraction, calculated by averaging the output `AOFlagger` flag array along the baseline axis. Bottom left: `SSINS` flagging fraction. Bottom right: χ^2 flagging fraction.

This does not immediately imply that per-baseline flagging should be abandoned in favor of per-observation flagging. The key question is whether the faint RFI missed by per-baseline flagging is **strong enough to bias an EoR power spectrum analysis**.

Consequences for power spectrum analyses

Measuring the 21-cm **power spectrum** is the main focus of most current-generation instruments targeting the EoR, including the MWA. To test the effects of faint, unflagged RFI on the power spectrum, we **inject simulated moving-RFI events** into otherwise clean MWA data and **compare several mitigation scenarios**:

1. Leaving the RFI unflagged.
2. Flagging the contaminated region on all baselines.
3. Applying the per-baseline `AOFlagger` algorithm.

This comparison struggles to separates two sources of power spectrum bias: residual contamination from missed RFI, and data loss from flagging. Preliminary results indeed suggest that, in small datasets, the loss of uv-coverage caused by flagging can itself introduce substantial power spectrum bias, sometimes **comparable to or larger** than the bias from residual unflagged RFI. This means that more aggressive flagging is not automatically better: removing contaminated data reduces RFI, but it also changes the sampling of the uv-plane.

Future work will increase both the observing volume and the number of injected RFI events. This will make it possible to isolate the power spectrum bias caused *specifically* by faint, partially flagged moving RFI, and to determine when baseline-aggregated flagging is preferable to per-baseline flagging.

References:

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